

2023 AMC 10B Problem 23 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10B Problem 23. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10B Problem 23

Problem:

An arithmetic sequence of positive integers has $n \geq 3$ terms, initial term a , and common difference $d > 1$. Carl wrote down all the terms in this sequence correctly except for one term, which was off by 1. The sum of the terms he wrote down was 222. What is $a + d + n$?

Answer Choices:

- A. 24
- B. 20
- C. 22
- D. 28
- E. 26

Solution:

Let the average value of the correct terms in the arithmetic sequence be r , which must be an integer or a half-integer. The sum of the terms is nr , which is either 221 or 223. The product $2nr$ is either $442 = 2 \cdot 13 \cdot 17$ or $446 = 2 \cdot 223$, so $\{n, 2r\}$ is one of $\{1, 442\}$, $\{2, 221\}$, $\{13, 34\}$, $\{17, 26\}$, $\{1, 446\}$, or $\{2, 223\}$. Because $n \geq 3$ and $2r > 2$, only $\{13, 34\}$ and $\{17, 26\}$ are possibilities. Furthermore, the least term in the sequence is $r - \frac{1}{2}(n-1)d$, which must be positive. This is the case if and only if $n = 13$, $r = 17$, $d = 2$, and $a = r - (n-1) = 5$. The requested sum is $a + d + n = 5 + 2 + 13 = \text{(B)} \boxed{20}$.

OR

The sum of the correct terms is

$$na + d \left(\frac{n(n-1)}{2} \right) = n \left(a + \frac{d(n-1)}{2} \right)$$

This sum is equal to either $221 = 13 \cdot 17$ or 223 , which is prime. But $n \geq 3$ divides the sum of the terms, so the sum is not prime and $n \left(a + \frac{d(n-1)}{2} \right) = 221$. Therefore $n = 13$ or $n = 17$.

If $n = 13$, then $a + \frac{d(n-1)}{2} = a + 6d = 17$. The only integer solution with $a > 0$ and $d > 1$ is $a = 5$ and $d = 2$. If $n = 17$, then $a + 8d = 13$. There are no integer solutions with $a > 0$ and $d > 1$ in this case.

The requested sum is $5 + 2 + 13 = \text{(B)} \boxed{20}$.

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